

# Daily Logic Drill #7

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

*New toolkit today: Mixed Synthesis Capstone · Every technique from Days 1–6, combined.*

*Attempt each question before checking answers. Time yourself. No calculators.*

## Section A Rapid Recognition

(≤15 s each)

*One quick question per technique from the whole week. Pure recognition, full speed.*

1. State the contrapositive of: “If a sequence is convergent, then it is bounded.”
2. What is the negation of: “No integer is both even and odd”?
3. Is “ $n$  is a multiple of 6” necessary, sufficient, both, or neither for “ $n$  is a multiple of 2”?
4. To prove “there is no largest prime number” by contradiction, what should you assume first?
5. Give a counterexample to: “Every odd number is prime.”
6. Name the fallacy: “If it is raining, the ground is wet. The ground is wet. Therefore it is raining.”
7. What is the converse of “If  $p$  then  $q$ ”?
8. To prove directly that “if  $n$  is a multiple of 7, then  $n^2$  is a multiple of 49”, what is the very first line of the proof?
9. True or false: “ $\exists x \forall y (x < y)$  and  $\forall y \exists x (x < y)$  are logically equivalent statements about real numbers.”
10. Which proof technique is most natural for disproving “every multiple of 3 is odd”?

## Section B Manipulation Drills

(≤50 s each)

*Each question pairs two techniques from different days.*

1. Prove by contrapositive: “If  $n$  is not a multiple of 3 and not a multiple of 5, then  $n$  is not a multiple of 15.”
2. Prove by contradiction: “There is no positive integer  $n$  such that  $n^2 + n$  is odd.”
3. Prove by induction:  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$  for all positive integers  $n$ .

4. A student wants to prove “if  $n$  is not a multiple of 5, then  $n^2$  is not a multiple of 25” by instead showing “if  $n$  is a multiple of 5, then  $n^2$  is a multiple of 25.” Is this a valid proof of the original claim? Justify your answer.
5. Give a counterexample showing that “ $n$  is prime” is not sufficient for “ $n$  is odd.”
6. Negate the statement: “For every set  $A$ , there exists a set  $B$  such that  $A \cap B = \emptyset$ .” Is the negated statement true or false?
7. Prove by induction that  $n^3 + 2n$  is divisible by 3 for all positive integers  $n$ .
8. Prove by contradiction that  $\sqrt{6}$  is irrational.
9. For a polynomial  $f$ , classify “ $f(x) > 0$  for all real  $x$ ” as necessary, sufficient, both, or neither for “ $f$  has no real roots.”
10. A student proves  $P(n)$  for all  $n \geq 1$  by checking  $P(1)$  directly, then showing  $P(k) \implies P(k+1)$  only for  $k \geq 2$  (skipping  $k = 1$ ). Explain the gap this leaves in the proof.

**Section C Substitution & Structure**( $\leq 90$  s each)*Tricky synthesis MCQs, combining techniques from across the week.*

1. Let  $P$ : “ $n$  is divisible by 4” and  $Q$ : “ $n$  is divisible by 2 and  $n$  is divisible by 6”. Which of the following is true?
  - A)  $P \iff Q$ .
  - B)  $P$  is sufficient but not necessary for  $Q$ .
  - C)  $P$  is necessary but not sufficient for  $Q$ .
  - D)  $P$  and  $Q$  are neither necessary nor sufficient for each other — e.g.  $n = 8$  satisfies  $P$  but not  $Q$ , and  $n = 6$  satisfies  $Q$  but not  $P$ .
  - E)  $Q$  is impossible to satisfy.
2. A theorem states: “If every element of a finite set  $S$  of positive integers is even, then the sum of all elements of  $S$  is even.” A student, wanting to prove the converse, argues: “Take  $S = \{2, 4\}$ ; the sum is 6, which is even, and every element is indeed even, confirming the converse.” What is the flaw?
  - A) The theorem itself is false.
  - B) A single supporting example cannot establish a universal converse statement — e.g.  $S = \{1, 3\}$  has an even sum (4) despite containing no even elements at all, disproving the converse.
  - C)  $S = \{2, 4\}$  does not actually have an even sum.
  - D) The original theorem’s proof technique (not the converse) is what’s flawed.
  - E) There is no flaw; the converse is proved.

3. Which proof technique is most directly suited to establishing: “There exists a positive integer  $n$  such that  $n^2 + 3n + 1$  is not prime”?
- A) Induction.
  - B) Direct exhibition of a specific such  $n$  (e.g.  $n = 6$ :  $6^2 + 3(6) + 1 = 55 = 5 \times 11$ ).
  - C) Contradiction.
  - D) Contrapositive.
  - E) Strong induction.
4. A student claims: “ $n$  is a multiple of 9” is necessary and sufficient for “the digit sum of  $n$  is a multiple of 9”. They verify this only for  $n = 9, 18, 27$  and declare the claim proved. Evaluate this.
- A) The claim and the verification are both fully correct.
  - B) The claim is true (it is a standard divisibility rule), but checking three cases does not constitute a proof of a necessary-and-sufficient claim holding for every positive integer  $n$ .
  - C) The claim is false; digit sums have nothing to do with divisibility by 9.
  - D) The claim is true only for necessity, not sufficiency.
  - E) The claim is true only for sufficiency, not necessity.
5. What is the contrapositive of “If  $n^2$  is not divisible by 4, then  $n$  is odd”?
- A) If  $n$  is odd, then  $n^2$  is not divisible by 4.
  - B) If  $n$  is even, then  $n^2$  is divisible by 4.
  - C) If  $n^2$  is divisible by 4, then  $n$  is even.
  - D) If  $n$  is not odd, then  $n^2$  is not divisible by 4.
  - E) If  $n^2$  is divisible by 4, then  $n$  is odd.
6. A student claims  $P(n)$ : “ $n^2 + 5n + 2$  is even for every positive integer  $n$ ” is provable by induction: base case  $P(1)$ :  $1 + 5 + 2 = 8$  is even; inductive step: assume  $k^2 + 5k + 2$  is even; then  $(k + 1)^2 + 5(k + 1) + 2 = k^2 + 7k + 8 = (k^2 + 5k + 2) + 2k + 6$ , which is (even)+(even)=even, so  $P(k + 1)$  holds. Is this induction valid, and is the underlying claim true?
- A) The induction is valid, and the claim is true for every positive integer — confirmed directly too, since  $n^2 + 5n + 2 = n(n + 5) + 2$ , and  $n(n + 5)$  (a product of two integers of opposite parity, since 5 is odd) is always even.
  - B) The induction has a flaw in the algebra.
  - C) The claim is false — there exists some  $n$  for which  $n^2 + 5n + 2$  is odd.
  - D) The base case is wrong.
7. Consider positive integers  $m, n$ . Statement  $P$ : “For every prime  $p$ , if  $p \mid mn$  then  $p \mid m$  or  $p \mid n$ .” Statement  $Q$ : “There exists a prime  $p$  such that  $p \mid mn$  but  $p \nmid m$  and  $p \nmid n$ .” Which of the following is true?
- A)  $P$  and  $Q$  can both be true simultaneously.

- B)  $Q$  is the negation of  $P$ , and since  $P$  is a true theorem (Euclid's Lemma) for all  $m, n$ ,  $Q$  is always false.
- C)  $P$  is the negation of  $Q$ .
- D) Neither  $P$  nor  $Q$  is ever true.
- E)  $P$  is false in general.
8. A student claims: “ $x = 1$ ” is necessary and sufficient for “ $x^3 - 1 = 0$ ” (for real  $x$ ), reasoning: “Sufficient, because substituting  $x = 1$  gives 0. Necessary, because  $1^3 - 1 = 0$ , which confirms  $x = 1$  works.” Evaluate the necessity argument specifically.
- A) Both parts of the reasoning are fully correct as given.
- B) The “necessity” argument is circular: showing  $1^3 - 1 = 0$  just re-verifies sufficiency again (that  $x = 1$  satisfies the equation), and says nothing about whether  $x = 1$  is the ONLY solution. Necessity IS true here — via  $x^3 - 1 = (x - 1)(x^2 + x + 1)$ , where the quadratic factor has no real roots (discriminant  $1 - 4 = -3 < 0$ ) — just not established by the reasoning given.
- C) The necessity argument is flawed because  $x = -1$  is also a solution.
- D) Sufficiency is actually false.
- E) There is no correct way to establish necessity here.

## Section D Challenge

(SMC / BMO1 difficulty)

*Full capstone proofs, drawing on every technique from the week.*

- Long John Silverman has buried treasure at an integer point  $(x, y)$  (not necessarily positive). He records the two values  $x^2 + y$  and  $x + y^2$ ; these two recorded numbers are distinct from each other. Prove that no other integer point  $(x', y') \neq (x, y)$  can produce the same pair of recorded values (i.e. satisfy both  $x'^2 + y' = x^2 + y$  and  $x' + y'^2 = x + y^2$ ) — so the map pins down the treasure's location uniquely. (after BMO1 2010 Q6)
- Two bags contain  $m$  and  $n$  balls respectively ( $m, n > 0$ ). Two operations are allowed, applied any number of times in any order: (a) remove the same number of balls from each bag; (b) triple the number of balls in one (chosen) bag. (after BMO1 2012 Q4)  
Prove that if  $m - n$  is odd, it is impossible to ever empty both bags (i.e. reach  $(0, 0)$ ) using only these two operations.
- Prove that there do not exist positive integers  $a < b < c$  with  $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$  and  $a, b, c$  all even.
- Prove, using strong induction, that every positive integer  $n \geq 2$  can be written as a product of primes in exactly one way, up to the order of the factors (the uniqueness part of the Fundamental Theorem of Arithmetic) — assuming as given Euclid's Lemma: if a prime  $p$  divides a product  $ab$ , then  $p \mid a$  or  $p \mid b$ .
- Consider the statement (\*): “For all positive integers  $m, n$ :  $m$  and  $n$  are coprime if and only if there exist integers  $x, y$  (not necessarily positive) with  $mx + ny = 1$ ” (Bézout's Identity).  
(a) Prove the  $\Leftarrow$  direction of (\*) directly.  
(b) Taking the  $\Rightarrow$  direction of (\*) as given (it is normally proved via the Euclidean algorithm),

use (\*) to prove: if  $p$  is prime and  $p \nmid a$ , then there exist integers  $x, y$  with  $px + ay = 1$ ; hence deduce Euclid's Lemma (if  $p \mid ab$  then  $p \mid a$  or  $p \mid b$ ) as a consequence.

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*“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”*

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Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)